

Andres Perez

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EDUCATION

University of Notre Dame

Bachelor of Science in Computer Engineering | Minor: Engineering Corporate Practice | GPA: 3.7

Notre Dame, IN

Graduation Date: May 2028

WORK EXPERIENCE

University of Notre Dame

Student Researcher

Notre Dame, IN

Aug. 2024 – Present

- Assisting Prof. Chaoli Wang with research by data labeling image data and downloading GSV data of households, surveying heat dissipation in household.

PROJECTS

IrishSat's Pi Hat PCB

Notre Dame, IN

Designer

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino and H-bridge circuits, improving system efficiency and reducing assembly complexity.
- Created and corrected the generated Bill of Materials and Component Placement List files from KiCad to export to JLCPCB for manufacturing.

Live Sensor Data Visualization System

Long Beach, CA

Developer

- Developed a real-time data visualization interface in Python (Tkinter/Matplotlib) for serially streamed MPU9250 sensor data from an Arduino-compatible microcontroller.
- Engineered a modular project structure including serial communication controllers, GUI logic, and sensor data parsing to enable future reuse across embedded systems projects.
- Validated end-to-end system by integrating sensor readings, asynchronous serial data handling, and python threading for smooth user interaction.

Portfolio Website | andres-perez0.github.io

Long Beach, CA

Developer

- Developed a portfolio website with Astro and Tailwind CSS frameworks as the front end

LEADERSHIP AND ACTIVITIES

IrishSat's Gravitational Orbital Attitude Thermal Lab

Notre Dame, IN

Electronics Lead

Aug. 2024 – Present

- Appointed as incoming Electronics Lead to oversee embedded systems development and PCB design initiatives for the Helmholtz cage project, coordinating hardware-software integration efforts across the cubesat projects.
- Engineered a Python-based data acquisition pipeline for magnetometer readings from Arduino, converting raw serial outputs into MATLAB-compatible formats for post-processing and modeling.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.

Long Beach Unified School District

Carson, CA

Ailurus Industries Electrical Team

Aug. 2023 – Jun. 2024

- Collaborated with a team of 30+ students to design, manufacture, and construct modular robots capable of reconfiguration for versatile tasks, including artifact retrieval.
- Enhanced UPenn's SMORES-EP cubic model by integrating an ESP32-UWB microcontroller, implementing a bevel gear design for differential drives. Applied NASA soldering standards in designing both physical body and electrical layouts, maintaining a 5-inch cube constraint for each module.

TECHNICAL AND LANGUAGE SKILLS

Programming: C, C++, Python, Git, MATLAB

Skills: Technical Writing, Electronics Soldering

Tools: Microsoft Office Suite, KiCad, Vim, SolidWorks, Fusion 360

Language: English (Native), Spanish (Bilingual)